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Subject : AI and SC

ASSIGNMENT 1

Module 1:

- 1] a) AI in marketing: AI analyses customer behaviour and preferences to recommend products and services. It uses predictive analysis to provide personalized suggestions.
Example: Netflix.
- b) AI in banking: AI powered chatbots provides 24x7 customer support and answer customer queries instantly. They improve customer experience by reducing response time.
Example: HDFC Bank's EVA chatbot.
- c) AI in Gaming: AI creates intelligent game opponents that can learn and make strategic decisions. It improves gameplay by adapting to player actions.
Example: alphaGo
- d) AI in Space Exploration: AI is used in processing satellite images, monitoring spacecraft systems, and assisting astronauts.

2. • Performance measure: It defines how the success of the agent is judged. It includes cleanliness of the floor, efficiency of cleaning, minimal energy / battery consumption and reduced noise.

• environment: setting in which the agent operates and perceives. It includes the room, floor type, furniture, walls and dirt.

• Actuators: Components through which the agent acts on the environment. They include wheels for movement, brushes for scrubbing, and a suction motor.

• sensors: Components through which the agent perceives its environment. They include dirt detection sensors, bump sensors and cliff sensors.

3. Wumpus world is a 4x4 grid-based environment where an agent must find gold while avoiding the Wumpus and deadly pits, using only percepts.

PEAS:

→ Performance measure: +1000 for gold, -1000 for death, -1 per action, -100 for shooting arrow.

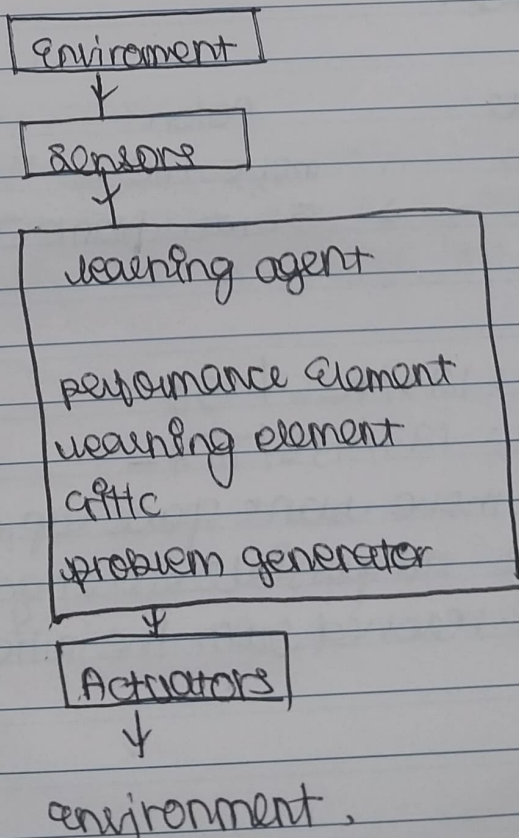
→ Environment: 4x4 grid containing the agent, Wumpus, gold and pits.

→ Actuators: move forward, left, right, grab gold, and shoot arrow.

→ Sensors: These are the components/percepts like, stench, breeze, glitter, bump and scream.

4. a) fully observable / partially observable: fully when the agent can access the complete state; otherwise partially observable.
 b) deterministic / stochastic: deterministic if the next state is completely determined by the current state; otherwise it is stochastic.
 c) episodic / sequential: episodic if each action is independent, sequential if current actions affect future decisions.
 d) static / dynamic: static if the environment does not change while the agent is deciding, dynamic if it does.

5. A learning agent is an AI agent that learns from its experiences and feedback and improves its performance overtime. It continuously updates its knowledge based on feedback to make better future decisions.



MODULE 2 :-

1. State space representation is a method of representing an AI problem using a set of all possible states of the problem and the actions/operators that can transform one state into another. It consists of:

- (i) Initial state: The starting condition of the problem
- (ii) Goal state: The desired final condition
- (iii) Operators / Actions: Actions that change one state into another
- (iv) State space: The collection of all possible states reachable from the initial state.

examples: 8-puzzle problem

In this, the board contains 8 numbered tiles and one blank space.

Initial state	Action	Goal state
1 2 3	move tile	1 2 3
4 _ 6	5 into blank →	4 5 6
7 5 8		7 8 _

→ Initial state: 123/4_6/758

→ Goal state: 123/456/78_

→ Operators: move blank space up, down, left, right.

→ State space: all possible arrangements of the tiles that can be reached from the initial state.

- 2] The water Jug Problem is a classical AI problem in which we have two jugs of fixed capacities and must measure a specific quantity of water using operations like fill, empty and pour.

Example :

consider two jugs of 4 litres and 3 litres. The goal is to obtain 2 litres of water in the 4-litre jug.

State space representation :

A state is represented as (X, Y) where :

- x = amount of water in 4L jug
- y = amount of water in 3L jug

Initial state : $(0, 0)$

Goal state : $(2, 0)$

Possible operations

- (i) Fill either jug completely
- (ii) Empty either jug completely
- (iii) Pour water from one jug into the other until the first is empty or the second is full.

PTO →

Solution:

Step	State (4L, 3L)	operation
1	(0,0)	Initial state
2	(4,0)	Fill 4L jug
3	(1,3)	Pour 4L $\rightarrow$ 3L
4	(1,0)	Empty 3L Jug
5	(0,1)	Pour 4L $\rightarrow$ 3L
6	(4,1)	Fill 4L jug
7	(2,3)	Pour 4L $\rightarrow$ 3L
8	(2,0)	Empty 3L Jug.

Therefore, 2 litres of water are obtained in the 4L jug.

5] Uninformed search methods are search techniques that have no additional information about the goal. They use only the initial state, operators, and goal test to explore the state space.

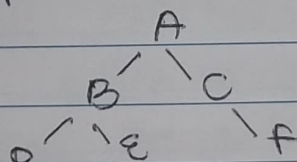
(i) Breadth first search (BFS)

$\rightarrow$ BFS explores the nodes level by level, starting from the root node.

$\rightarrow$ It uses a FIFO queue

$\rightarrow$ BFS is complete and gives the shortest path when all step costs are equal.

Example:

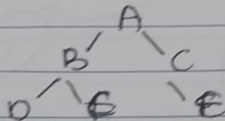


BFS order: $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F$

ii) Depth first search (DFS)

- DFS explores a path as deeply as possible before backtracking.
- It uses a LIFO stack or recursion
- DFS requires less memory but may not find the shortest path

Example:



DFS order: $A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F$

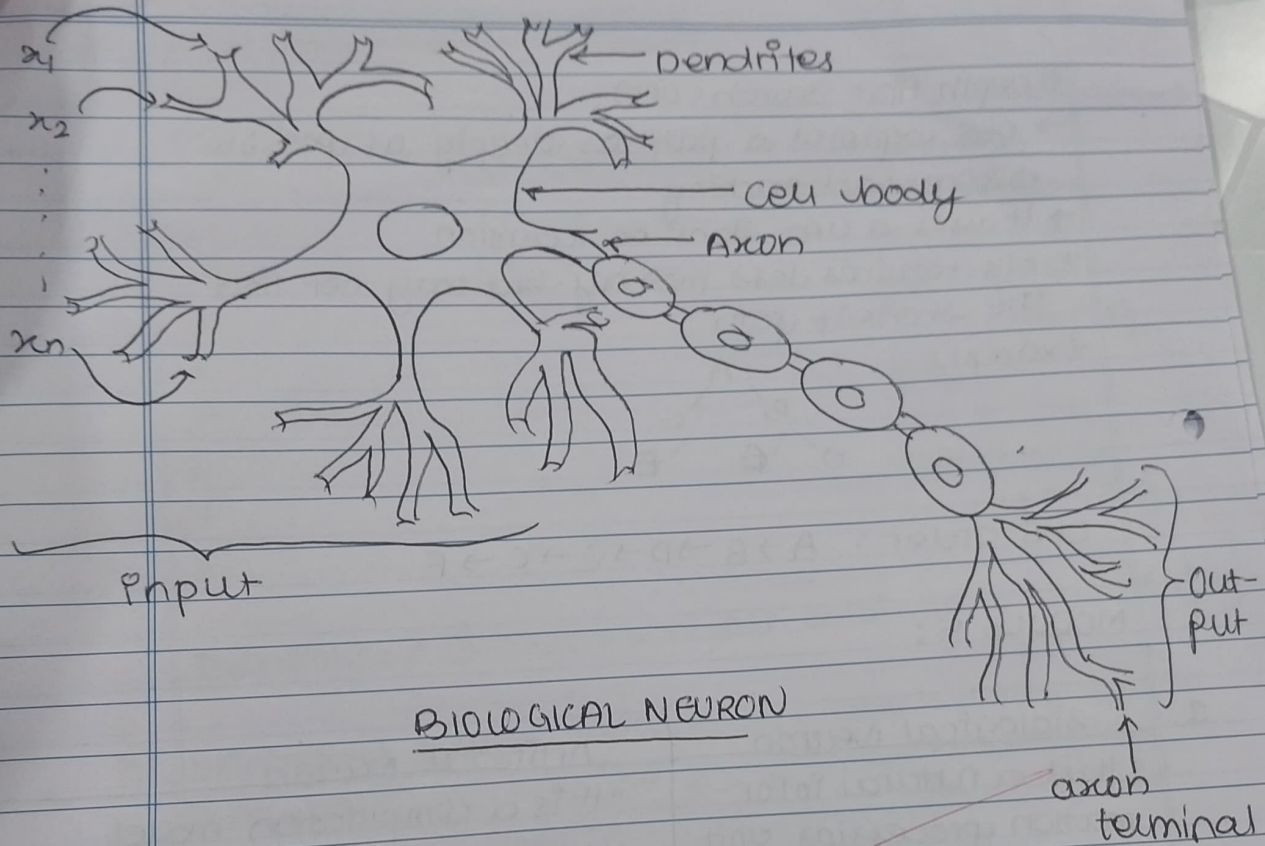
MODULE 5:

1. Biological Neuron

- It is a natural information processing unit found in human body
- consists of dendrite, cell body and axon
- Dendrites receive signals from other neurons
- communicates using electrochemical signals
- connections between neurons are formed through biological synapse.

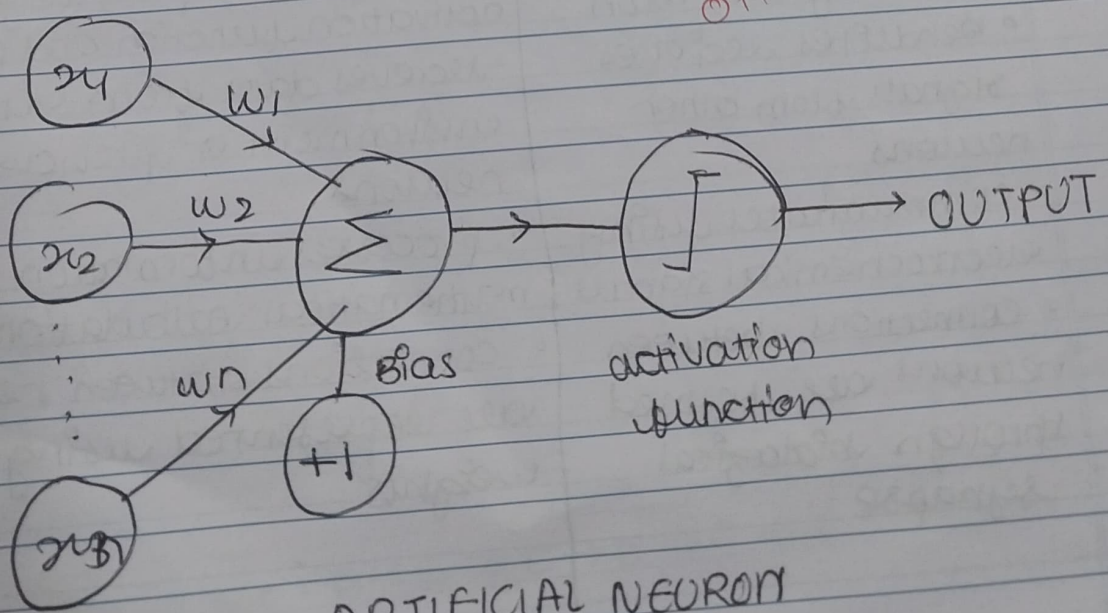
Artificial Neuron

- It is a computation model inspired by biological neurons
- consists of inputs, weights, bias, activation function and output.
- receives data from external environment or previous neurons
- processes information using mathematical calculations
- connections between neurons are represented using weights



BIOLOGICAL NEURON

Red signature
01/09/26



ARTIFICIAL NEURON